

Table 1: Some of the mice 2l imputation methods

Type	Package	Imputation method	Variable type	Description	Heteroscedasticity	Systematic missing	Predictors type*
1-stage	mice	2l.norm	Continuous (level-1)	Uses LMM and requires at least 1 observation per cluster (Kasim, 1998).	Yes	No	0, 2
		2l.lmer		Uses an LMM estimated with lme4 (Jolani, 2017).	Yes	Yes	0, 1, 2
		2l.pan		LMM with homogeneous within group variances, relying on the pan R package (Schafer, 1997).	No	Yes	-1, 0, 1, 2, 3, 4
		2l.bin	Binary (level-1)	Better for small size clusters. It fits a GLMM with link (logit) using lme4 (Jolani, 2015).	Yes	Yes	0, 1, 2
		2l.only. (mean, norm, pmm)	Continuous, binary, categorical (level-2)	It can be used to check if level-2 variables are consistent across clusters.		Yes, sporadic only with 2l.mean	0, 1
2-stage	miceadds	2l. (continuous, binary, pmm)	Continuous, binary, categorical (level-1)	GLMM estimated with lme4 or the bayesian blme package (Snijders, 2012; Zinn, 2013).		Yes	0, 1, 2, 3, 4
		2l.glm. (norm, bin, pois)	Continuous, binary, count (level-1)	Bayesian GLMM with non-informative prior (Jeffrey). Better for few clusters and few observations per cluster. Binary logit link. (Jolani, 2017; Audigier 2018).	No	Yes	0, 2
	Github repository	2l.jomo	Continuous, binary (level-1)	Bayesian MCMC approach, with a GMM based on conjugated priors (Quaragno Carpenter, 2016). Binary probit link. It relies on the jomo package.	Yes	Yes	0, 2
		heckman.quart	Binary, ordered (level-1)	MNAR based on Heckman model using quadrature estimation with random intercept (Hammon 2020, 2022).	Yes	Yes	0, 1
		2l.2stage. (norm, bin, pmm, pois)	Continuous, binary, categorical, count (level-1)	Better for large size clusters. (Resche-Rignon, White, 2016; Audigier 2018). Binary logit link.	Yes	Yes	0, 2
	micemd	2l.2stage.heckman	Continuous, binary (level-1)	MNAR based on Heckman model estimated in two stages	Yes	Yes	-4, -3, 0, 2

LMM: Linear mixed model, GLMM: Generalized linear mixed model. * See Box 4 of the manuscript.